
Exercise 8

1 Explicit MPC

Compute the explicit MPC controller for the following

$$\begin{aligned} u^*(x) = \operatorname{argmin} \quad & 2u^\top u + (x + u)^\top H(x + u) \\ \text{s.t.} \quad & u \in [-1, 1] \end{aligned}$$

where H is the solution to the discrete-time algebraic Riccati equation for $A = 1$, $B = 1$, $Q = 1$ and $R = 2$.

Compute H from the algebraic Riccati equation

$$\begin{aligned} H &= Q + A^\top H A - A^\top H B (R + B^\top H B)^{-1} B^\top H A \\ H &= 1 + H - \frac{H^2}{H + 2} \\ 0 &= H^2 - H - 2 = (H - 2)(H + 1) \end{aligned}$$

$H = 2 > 0$ is the solution, so the parametric optimization problem is now

$$\begin{aligned} u^*(x) = \operatorname{argmin} \quad & 2u^\top u + 2(x + u)^\top (x + u) \\ \text{s.t.} \quad & u \in [-1, 1] \end{aligned}$$

We solve for the case with no constraints by setting the derivative to zero:

$$4u^* + 4(x + u^*) = 0 \Rightarrow u = -\frac{x}{2}$$

This is the control law for $u \in [-1, 1]$. The control law will saturate if u is outside this range.

$$u^*(x) = \begin{cases} -1 & \text{if } x \in (2, \infty) \\ -\frac{x}{2} & \text{if } x \in [-2, 2] \\ 1 & \text{if } x \in (-\infty, -2) \end{cases}$$